

Machine Learning Notes (Modules 1–3)

These notes are concise and exam-oriented, suitable for KTU-style preparation.

Module 1: Overview of Machine Learning

What is Machine Learning?

Machine Learning (ML) is a branch of AI where computers learn patterns from data and make predictions or decisions without being explicitly programmed.

Example: An email spam filter learns from previous emails to identify spam.

Machine Learning Paradigms

1. Supervised Learning

- Trains using **labeled data** (input + correct output).
- Goal: Learn a mapping from inputs to outputs.

Applications:

- House price prediction
- Email spam detection
- Image classification

Algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Naive Bayes

Example:

Input	Output
House size	Price
Student hours studied	Exam marks

2. Unsupervised Learning

- Uses **unlabeled data**.

- Finds hidden structures or patterns.

Applications:

- Customer segmentation
- Data clustering
- Market basket analysis

Common algorithms:

- K-Means
 - Hierarchical Clustering
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3. Semi-Supervised Learning

- Combines **small labeled data + large unlabeled data**.
- Useful when labeling data is expensive.

Example:

Medical image classification with only a few expert-labeled scans.

4. Reinforcement Learning

- Agent learns by interacting with an environment.
- Receives **rewards** or **penalties**.
- Goal: Maximize cumulative reward.

Components:

- Agent
- Environment
- Action
- State
- Reward

Applications:

- Robotics
- Self-driving cars

- Chess/Go playing AI

Parameter Estimation

Maximum Likelihood Estimation (MLE)

Chooses parameter values that make the observed data **most likely**.

Formula

$$\begin{aligned} & [\\ & \hat{\theta}_{MLE} = \arg\max_{\theta} P(\text{Data}|\theta) \\ &] \end{aligned}$$

$$\hat{\theta}_{MLE} = \arg\max_{\theta} P(\text{Data} \mid \theta)$$

Intuition

Given several possible parameter values, pick the one that best explains the observed data.

Advantages

- Simple
- Widely used
- Works well with large datasets

Maximum A Posteriori (MAP)

Uses both observed data **and prior knowledge**.

Formula

$$\begin{aligned} & [\\ & \hat{\theta}_{MAP} = \arg\max_{\theta} P(\theta|\text{Data}) \\ &] \end{aligned}$$

$$\hat{\theta}_{MAP} = \arg\max_{\theta} P(\theta \mid \text{Data})$$

Using Bayes' theorem:

$$\begin{aligned} & [\\ & P(\theta|\text{Data}) \propto P(\text{Data}|\theta)P(\theta) \\ &] \end{aligned}$$

Difference between MLE and MAP

MLE

Uses only observed data

Maximizes likelihood

No prior information

MAP

Uses observed data + prior belief

Maximizes posterior probability

Prior information included

Bayesian Formulation

Based on **Bayes' Theorem**:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

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Where:

- **Prior** = Initial belief
- **Likelihood** = Probability of data given hypothesis
- **Posterior** = Updated belief after seeing data
- **Evidence** = Normalizing factor

Bayesian methods continuously update predictions as new data arrives.

Module 2: Supervised Learning

Regression

Regression predicts **continuous numerical values**.

Example:

- Salary prediction
- Temperature prediction
- House price estimation

Linear Regression (One Variable)

Relationship between one input and output:

$$y = mx + b$$

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Where:

- m = slope
- b = intercept

Example:

Predict house price from area.

Linear Regression (Multiple Variables)

Uses multiple input features.

$$y = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

Useful when prediction depends on many factors such as area, bedrooms, and age.

Gradient Descent

An optimization algorithm used to minimize prediction error.

Steps

1. Initialize parameters randomly.
2. Compute predictions.
3. Calculate cost/error.
4. Compute gradients.
5. Update parameters.
6. Repeat until convergence.

Learning rate (α):

- Too large $\rightarrow$ overshoots minimum.

- Too small → converges slowly.
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Matrix Method (Normal Equation)

Instead of iterative updates, solve directly:

$$\theta = (X^T X)^{-1} X^T y$$

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Advantages:

- Exact solution
- No learning rate required

Disadvantage:

- Computationally expensive for very large datasets.
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Overfitting

Occurs when a model memorizes training data instead of learning general patterns.

Symptoms

- Very high training accuracy
- Poor performance on unseen data

Prevention

- More training data
 - Regularization
 - Simpler models
 - Cross-validation
-

Linear Methods for Classification

Logistic Regression

Used for **binary classification**.

Examples:

- Spam vs Not Spam
- Pass vs Fail
- Disease vs Healthy

Uses the sigmoid function:

[
 $\sigma(z) = \frac{1}{1 + e^{-z}}$
]

$\sigma(z) = \frac{1}{1 + e^{-z}}$

Output lies between **0 and 1**, representing probability.

Naive Bayes

Based on Bayes' theorem and assumes features are conditionally independent.

Advantages

- Fast
- Works well for text classification
- Performs well on small datasets

Applications

- Spam detection
- Sentiment analysis
- Document classification

Decision Tree (ID3 Algorithm)

A tree-based classifier where:

- Root = starting decision
- Internal nodes = attribute tests
- Leaves = class labels

ID3 selects attributes using Information Gain.

Process

1. Compute entropy.

2. Calculate information gain for each attribute.
3. Choose the attribute with highest gain.
4. Split the data.
5. Repeat recursively.

Advantages

- Easy to interpret
 - Handles categorical data well
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Module 3: Neural Networks and Support Vector Machines

Perceptron

The simplest neural network model for binary classification.

Components

- Inputs
- Weights
- Weighted sum
- Activation function
- Output

Decision rule:

```
[  
y=  
\begin{cases}  
1,&\text{if } wx+b>0\  
0,&\text{otherwise}  
\end{cases}  
]
```

It can correctly classify only **linearly separable** data.

Multilayer Feed Forward Neural Network (MLP)

Contains:

1. Input layer

2. One or more hidden layers
3. Output layer

Information flows only in the forward direction (no cycles).

Advantages

- Learns complex nonlinear relationships
 - Used in image recognition, NLP, and speech processing
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Activation Functions

1. Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- Output: 0 to 1
 - Smooth and differentiable
 - Used for probabilities
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2. ReLU (Rectified Linear Unit)

$$f(x) = \max(0, x)$$

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- Fast computation
 - Avoids saturation for positive values
 - Most widely used in deep learning
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3. Tanh

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

- Output range: -1 to 1

- Zero-centered
 - Often performs better than sigmoid in hidden layers
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Backpropagation Algorithm

Backpropagation trains neural networks by updating weights to reduce prediction error.

Steps

1. Forward propagation (compute output).
2. Calculate loss/error.
3. Propagate error backward through layers.
4. Compute gradients.
5. Update weights using gradient descent.
6. Repeat until convergence.

Purpose: Minimize the network's prediction error and improve accuracy.

Support Vector Machine (SVM)

SVM is a supervised learning algorithm mainly used for **classification** (and sometimes regression).

Key Idea

Find the **optimal hyperplane** that separates classes with the **maximum margin**.

- **Support vectors:** Data points closest to the decision boundary.
- **Margin:** Distance between the hyperplane and nearest data points.
- Larger margin generally improves generalization.

Advantages

- Effective in high-dimensional spaces
- Works well with small and medium-sized datasets
- Can model nonlinear boundaries using kernels

Common Applications

- Face recognition

- Text classification
- Bioinformatics
- Handwriting recognition

Quick Exam Revision

Topic	Key Point
Supervised Learning	Uses labeled data
Unsupervised Learning	Uses unlabeled data
Semi-Supervised Learning	Mix of labeled and unlabeled data
Reinforcement Learning	Learns through rewards and penalties
MLE	Maximizes likelihood ($P(\text{Data} \mid \theta)$)
MAP	Maximizes posterior using prior knowledge
Linear Regression	Predicts continuous values
Logistic Regression	Binary classification using sigmoid
Gradient Descent	Iteratively minimizes error
Overfitting	Excellent on training data, poor on new data
Naive Bayes	Bayes theorem with independence assumption
ID3	Builds decision trees using information gain
Perceptron	Single-layer linear classifier
MLP	Neural network with hidden layers
Sigmoid	Outputs values in $([0, 1])$
ReLU	Outputs $\max(0, x)$
Tanh	Outputs values in $([-1, 1])$
Backpropagation	Updates weights using error gradients
SVM	Finds a maximum-margin separating hyperplane

